

Task 1 Report (Cashier Program)

Student Information:

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Project: Cashier Program for Future Mall

Objective:

The objective of this project is to develop a simple cashier system using Python. The program allows the cashier to enter customer purchases, calculates the total bill, applies a discount when required, and prints the final receipt.

Project Requirements:

The program includes the following features:

- Create a list containing five product names.
- Create another list containing the prices of the products.
- Display all available products.
- Allow the cashier to enter products using a loop.
- Calculate the total price.
- Apply a 10% discount if the total is greater than 500 EGP.
- Print the final receipt with the mall name.
- Print a thank-you message.

How the Program Works:

The program starts by storing the mall name, product names, and product prices in variables and lists.

After displaying the available products, the cashier enters the product number.

The program checks whether the entered number is valid.

If the product exists, its price is added to the total bill.

The cashier can continue adding products until entering **n**.

Finally, the program checks whether the total exceeds 500 EGP. If the condition is true, a 10% discount is applied before printing the final receipt.

Functions Used:

print_all()

This function displays all products and their prices.

final()

This function prints the final receipt including:

- Mall name
- Total price
- Discount
- Final total
- Thank you message

Programming Concepts Used:

The project uses the following Python concepts:

- Variables
- Lists
- Functions
- While Loop
- If-Else Statements
- User Input
- Arithmetic Operations
- String Formatting (f-string)

Screenshots:

```
StudentID-Mall.py X
StudentID-Mall.py > ...
1  mall_name = "Future Mall"
2
3  products = [
4      "Milk",
5      "Bread",
6      "Rice",
7      "Apple",
8      "Juice"
9  ]
10
11  prices = [
12      60,
13      25,
14      120,
15      40,
16      90
17  ]
18
19  def print_all():
20      for i in range(5):
21          print(f'-{i} {products[i]} ==> {prices[i]} EG')
22
23  print_all()
24
25  total = 0
26
27  while True:
28      products_number = int(input('Enter the Product-number: '))
29      if products_number in range(5):
30          price = prices[products_number]
31          total += price
32      else:
33          print('Sorry, This Product is not deffind!')
34
35      chick = input('Add another product? (y,n): ').strip()
36      if chick.lower() == 'y':
37          continue
38      else:
39          break
40
41  if total > 500:
42      discount = total * 0.10
43      final_total = total - discount
44  else :
45      discount = 0
```

```

StudentID-Mall.py > ...
24
25     total = 0
26
27     while True:
28         products_number = int(input('Enter the Product-number: '))
29         if products_number in range(5):
30             price = prices[products_number]
31             total += price
32         else:
33             print('Sorry, This Product is not deffind!')
34
35         chick = input('Add another product? (y,n): ').strip()
36         if chick.lower() == 'y':
37             continue
38         else:
39             break
40
41     if total > 500:
42         discount = total * 0.10
43         final_total = total - discount
44     else :
45         discount = 0
46         final_total = total
47
48
49     def final():
50         print('='*15 , mall_name , '='*15)
51         print(f'- Total: {total}')
52         print(f'- Discount: {discount}')
53         print(f'- Final Total: {final_total}')
54         print('='*10 , 'Thank you for shopping!' , '='*10)
55
56
57     final()

```

```

C:\Users\Abdallahrahan Taha\Documents\Final Project DICE\Graduation_Project>python StudentID-Mall.py
-0 Milk ==> 60 EG
-1 Bread ==> 25 EG
-2 Rice ==> 120 EG
-3 Apple ==> 40 EG
-4 Juice ==> 90 EG
Enter the Product-number: 0
Add another product? (y,n): y
Enter the Product-number: 1
Add another product? (y,n): y
Enter the Product-number: 3
Add another product? (y,n): y
Enter the Product-number: 2
Add another product? (y,n): y
Enter the Product-number: 4
Add another product? (y,n): n
===== Future Mall =====
- Total: 335
- Discount: 0
- Final Total: 335
===== Thank you for shopping! =====

```

```

C:\Users\Abdalrahman Taha\Documents\Final Project DICE\Graduation_Project>python StudentID-Mall.py
-0 Milk ==> 60 EG
-1 Bread ==> 25 EG
-2 Rice ==> 120 EG
-3 Apple ==> 40 EG
-4 Juice ==> 90 EG
Enter the Product-number: 0
Add another product? (y,n): y
Enter the Product-number: 0
Add another product? (y,n): y
Enter the Product-number: 0
Add another product? (y,n): y
Enter the Product-number: 1
Add another product? (y,n): y
Enter the Product-number: 1
Add another product? (y,n): y
Enter the Product-number: 1
Add another product? (y,n): y
Enter the Product-number: 2
Add another product? (y,n): y
Enter the Product-number: 2
Add another product? (y,n): y
Enter the Product-number: 2
Add another product? (y,n): y
Enter the Product-number: 2
Add another product? (y,n): y
Enter the Product-number: 2
Add another product? (y,n): y
Enter the Product-number: 4
Add another product? (y,n): y
Enter the Product-number: 4
Add another product? (y,n): y
Enter the Product-number: 2
Add another product? (y,n): n
===== Future Mall =====
- Total: 1155
- Discount: 115.5
- Final Total: 1039.5
===== Thank you for shopping! =====

```

Conclusion:

This project successfully implements a simple cashier system using Python.

The program allows the cashier to enter purchases, calculates the total price, applies a 10% discount when the total exceeds 500 EGP, and prints a clear final receipt.

The project demonstrates the use of Python fundamentals such as lists, loops, functions, conditions, and user input.